

Chapter 8 - Vision-Language Models with CLIP

Exam Study Guide with Symbolic Equations, Sample Questions, and De-duplicated Flashcards

Source focus	Chapter 8 lecture, Chapter 8 summary/sample questions, and Chapter 8 flashcards.
Main exam goal	Understand VLM families, CLIP contrastive learning, similarity matrices, temperature, pull-push gradients, zero-shot classification, retrieval, fusion/generative VLMs, and distillation.
Equation style	Equations use mathematical symbols such as τ , Σ , $\ \cdot\ _2$, $\hat{v}$, $\hat{t}$, $L_{\{I \rightarrow T\}}$, and $p(y_t y_{<t}, x)$ rather than word-only descriptions.

Main Exam Topics

1. What a Vision-Language Model learns

A Vision-Language Model (VLM) connects visual semantics such as objects, actions, and relations with linguistic semantics such as words, phrases, and sentences.

The key goal is not only image classification or text generation alone. The goal is to make images and text comparable, searchable, and usable together.

This enables multimodal tasks such as text-to-image retrieval, image-to-text retrieval, zero-shot recognition, caption generation, visual question answering, grounding, and instruction following.

$D = \{(x_i, y_i)\}_{i=1}^N$, where x_i is an image and y_i is paired text

2. The two dominant VLM architecture families

Dual-encoder contrastive models, such as CLIP, use separate image and text encoders. They map both modalities into a shared embedding space and compare them with a similarity score.

Fusion or encoder-decoder models combine visual and text tokens through cross-attention or a shared Transformer. They are more natural for captioning, VQA, and instruction-following because each output token can consult the image.

The exam distinction is important: dual encoders are efficient for retrieval and zero-shot matching, while fusion models are stronger for detailed conditional generation and grounding.

Dual encoder: $v = f_{\{\theta I\}}(x)$, $t = g_{\{\theta T\}}(y)$

Fusion decoder: $p(y_t | y_{<t}, x)$

3. CLIP embeddings, normalization, and similarity

CLIP uses an image encoder and a text encoder. Their outputs are usually L2-normalized so that dot products become cosine similarities.

Normalization makes comparison depend mainly on direction in semantic space rather than raw vector magnitude.

The similarity score is often divided by a temperature τ , or equivalently multiplied by a learned logit scale $1/\tau$.

$v_i = f_{\{\theta I\}}(x_i)$, $t_j = g_{\{\theta T\}}(y_j)$

$\hat{v}_i = v_i / \|v_i\|_2$, $\hat{t}_j = t_j / \|t_j\|_2$

$$s_{\{ij\}} = \hat{v}_i^T \hat{t}_j / \tau$$

4. Symmetric contrastive loss

In a batch of B matched image-text pairs, CLIP compares every image with every text. The diagonal pairs are positives, and the off-diagonal pairs are batch negatives.

The image-to-text loss asks: given an image, which text in the batch is the correct match? The text-to-image loss asks the symmetric question.

The final CLIP loss averages both directions, forcing alignment from image to text and from text to image.

$$p_{\{ij\}}^{\{I \rightarrow T\}} = \exp(s_{\{ij\}}) / \sum_{k=1}^B \exp(s_{\{ik\}})$$

$$L_{\{I \rightarrow T\}} = -(1/B) \sum_{i=1}^B \log p_{\{ii\}}^{\{I \rightarrow T\}}$$

$$L = (L_{\{I \rightarrow T\}} + L_{\{T \rightarrow I\}}) / 2$$

5. Pull-push gradient intuition

The contrastive gradient has a useful geometry. It pulls the image embedding toward its matched positive text and pushes it away from a weighted mixture of negative texts.

The hardest negatives get the largest weights because their current similarity scores are high. This is why contrastive learning teaches not only what matches, but also what should not match.

Temperature controls how sharply the model focuses on hard negatives: smaller τ gives a peakier softmax and stronger hard-negative emphasis.

$$\partial l_i / \partial \hat{v}_i = (1/\tau) (\sum_{j=1}^B p_{\{ij\}} \hat{t}_j - \hat{t}_i)$$

$$\text{update direction} \propto \hat{t}_i - \sum_j p_{\{ij\}} \hat{t}_j$$

6. Zero-shot classification with CLIP

CLIP performs zero-shot classification by turning class names into text prompts, embedding those prompts, embedding the image, and choosing the class with the highest similarity.

This avoids a conventional classifier head trained on the target classes. The labels are introduced at test time through natural language.

Prompt wording matters. Prompt ensembling averages multiple prompt templates to make the class prototype more robust.

$$w_c = (1/K) \sum_{k=1}^K \text{normalize}(g(\text{prompt}_k(c)))$$

$$\hat{c} = \text{argmax}_c \hat{v}^T w_c / \tau$$

7. Retrieval with dual encoders

Dual encoders are efficient because image embeddings and text embeddings can be computed independently and cached.

For text-to-image retrieval, embed the text query and rank images by similarity. For image-to-text retrieval, embed the image query and rank texts by similarity.

At large scale, Approximate Nearest Neighbor (ANN) search is commonly used to retrieve top-k items quickly.

$$\text{text} \rightarrow \text{image: retrieve } x_j \text{ with } \max_j \hat{v}_j^T \hat{t}_q$$

$$\text{image} \rightarrow \text{text: retrieve } y_j \text{ with } \max_j \hat{v}_q^T \hat{t}_j$$

$$\cos(A, B) = A^T B / (\|A\|_2 \|B\|_2)$$

8. Fusion VLMs and cross-attention

Fusion models combine visual and text information inside the model rather than only comparing final global embeddings.

In an encoder-decoder VLM, the image encoder produces visual memory M . The decoder uses masked self-attention over the text prefix and cross-attention into the visual memory.

This makes fusion models stronger for captioning, VQA, and detailed grounding because individual generated words can attend to image regions or visual tokens.

$$M = \text{ImageEncoder}(x)$$
$$\text{CrossAttention}(Q_{\text{text}}, K_M, V_M) = \text{softmax}(Q_{\text{text}} K_M^T / \sqrt{d_k}) V_M$$
$$\log p(y|x) = \sum_{t=1}^T \log p(y_t | y_{<t}, x)$$

9. Teacher forcing, causal masks, and LM head

Generative VLMs train the decoder with teacher forcing: the decoder receives the ground-truth previous caption tokens during training.

A causal mask prevents text tokens from looking at future tokens. This preserves the autoregressive meaning of the conditional generation objective.

The LM head maps decoder hidden states into vocabulary logits so the model can predict the probability of the next word or token.

$$L_{\text{gen}} = -\sum_{t=1}^T \log p_{\psi}(y_t | y_{<t}, x)$$
$$\text{logits}_t = W_{\text{LM}} h_t + b_{\text{LM}}$$
$$p(y_t | y_{<t}, x) = \text{softmax}(\text{logits}_t)$$

10. Data quality, failure modes, and evaluation

VLMs are often trained on web-scale paired image-text data. These pairs are powerful but noisy: alt-text may be incomplete, biased, irrelevant, or inaccurate.

Important failure modes include hallucination, weak grounding, bias and spurious correlations, and distribution shift to specialized domains.

Evaluation should include more than one benchmark: retrieval Recall@K, zero-shot accuracy, robustness, fairness/bias, latency, memory, and energy can all matter.

$$\text{Recall@K} = (\# \text{ queries with correct item in top K}) / (\# \text{ queries})$$

top-k retrieval usually ranks by cosine similarity

11. Variants, adaptation, and efficient VLMs

SigLIP modifies the contrastive objective by replacing batch-wide softmax with a pairwise sigmoid loss. CoOp replaces hand-written prompts with learnable prompt vectors.

OWL-ViT and related methods extend image-text alignment toward dense prediction and localization. LLaVA-style models add visual instruction tuning for dialogue about images.

TinyCLIP and MobileCLIP focus on efficient deployment. Distillation trains a small student to mimic a large teacher, often using both embedding regression and similarity-matrix KL losses.

$$L_{\text{embed}} = \|z_{\text{student}} - z_{\text{teacher}}\|_2^2$$
$$L_{\text{KL}} = \text{KL}(\text{softmax}(S_{\text{teacher}}/\tau_d) \parallel \text{softmax}(S_{\text{student}}/\tau_d))$$

Quick Formula Sheet

Concept	Symbolic equation / meaning
Paired data	$D = \{(x_i, y_i)\}_{i=1}^N$
Image and text encoders	$v_i = f_{\theta_I}(x_i), t_j = g_{\theta_T}(y_j)$
L2 normalization	$\hat{v}_i = v_i / \ v_i\ _2, \hat{t}_j = t_j / \ t_j\ _2$
Cosine similarity	$\cos(A, B) = A^T B / (\ A\ _2 \ B\ _2)$
CLIP score matrix	$s_{ij} = \hat{v}_i^T \hat{t}_j / \tau$
Image-to-text softmax	$p_{ij} = \exp(s_{ij}) / \sum_{k=1}^B \exp(s_{ik})$
Image-to-text loss	$L_{I \rightarrow T} = -(1/B) \sum_i \log p_{ii}$
Symmetric CLIP loss	$L = (L_{I \rightarrow T} + L_{T \rightarrow I}) / 2$
Pull-push gradient	$\partial \ell_i / \partial \hat{v}_i = (1/\tau)(\sum_j p_{ij} \hat{t}_j - \hat{t}_i)$
Zero-shot class prototype	$w_c = (1/K) \sum_k \text{normalize}(g(\text{prompt}_k(c)))$
Zero-shot prediction	$\hat{c} = \text{argmax}_c \hat{v}^T w_c / \tau$
Conditional generation	$\log p(y x) = \sum_t \log p(y_t y_{\leq t}, x)$
Cross-attention	$\text{softmax}(Q_{\text{text}} K_M^T / \sqrt{d_k}) V_M$
Temperature/logit scale	$\text{logit_scale} = 1/\tau \Rightarrow \tau = 1/\text{logit_scale}$

Sample Exam Questions with Answers

Question 1

Why is CLIP training fundamentally about both positive and negative pairs?

Sample answer

CLIP must increase similarity for matched image-text pairs while decreasing similarity for mismatched pairs. If it only pulled matched pairs together, unrelated items could also collapse into the same region. Batch negatives shape the geometry of separation and teach the model which distinctions matter.

Question 2

Why is temperature more than a cosmetic scaling constant in CLIP?

Sample answer

The temperature τ controls the sharpness of the softmax over candidate matches. A smaller τ makes the distribution peakier and emphasizes hard negatives more strongly, while a larger τ smooths the distribution and weakens the pressure to separate close competitors.

Question 3

Why does zero-shot classification not require a conventional classifier head?

Sample answer

CLIP reframes classification as similarity matching. The image is compared to text prompt embeddings for class names, and the class with the highest similarity is selected. Therefore, the class meanings enter through language prompts rather than a trained task-specific output layer.

Question 4

What is the conceptual difference between masked self-attention and cross-attention in a fusion VLM?

Sample answer

Masked self-attention organizes the previously generated text while preserving the autoregressive constraint. Cross-attention lets the decoder consult visual memory from the image encoder, so the current text token can use image evidence.

Question 5

Why is teacher forcing useful, and what is its weakness?

Sample answer

Teacher forcing stabilizes training by feeding the correct previous caption tokens to the decoder. Its weakness is exposure bias: at inference time the model conditions on its own generated tokens, so early mistakes can affect later predictions.

Flashcards - All Unique Questions

The flashcard section keeps the full Chapter 8 review set while cleaning repeated or OCR-noisy wording. Use the question column for active recall, then check the answer column.

#	Question	Answer
1	What is the primary function of Vision-Language Models (VLMs)?	They connect visual semantics such as objects/actions with linguistic semantics such as words/sentences across modalities.
2	In a dual-encoder VLM like CLIP, how are the two modalities compared?	By mapping image and text into a shared embedding space and computing their cosine similarity.
3	What is the purpose of L2-normalization in CLIP architecture?	It projects embeddings onto a unit hypersphere so the dot product directly represents cosine similarity.
4	In CLIP, the similarity score between image x and text y is $s(x,y)=v^T t$. What do v and t represent?	They are the normalized feature vectors produced by the image encoder and text encoder, respectively.
5	What role does the temperature parameter τ play in the CLIP contrastive loss?	It scales the logits before the softmax and controls the peakiness of the probability distribution.
6	What is the intuition behind a very small temperature value τ in contrastive learning?	It makes the model more sensitive to similarity differences and forces stronger focus on the hardest negatives.
7	In the contrastive loss function, what is the goal of the image-to-text loss $L_{I \rightarrow T}$?	To maximize the probability that an image matches its correct caption among a batch of possible texts.
8	How is the symmetric loss calculated in CLIP training?	It is the average of the image-to-text cross-entropy loss and the text-to-image cross-entropy loss.
9	According to the pull-push geometry of CLIP gradients, what does pull mean?	It means moving the image embedding v_i closer to its matched positive text embedding t_i .
10	In the CLIP update rule, what determines which negatives are pushed away most strongly?	Negatives with the highest current similarity scores, called hard negatives, receive stronger push-away gradients.
11	What is the mechanism used by CLIP for zero-shot classification?	The model treats text prompts of class names as labels and chooses the class prompt with the highest cosine similarity to the image.
12	Term: Prompt Ensembling	Averaging embeddings from multiple text templates, such as "a photo of a {}" and "an image of a {}", to improve classification robustness.
13	Why is CLIP referred to as an open-vocabulary recognition model?	It can classify images using arbitrary text strings rather than being restricted to a fixed predefined label set.
14	What is the main architectural difference between a dual-encoder VLM and a fusion VLM?	Dual encoders use separate streams for alignment, while fusion models mix visual and text tokens inside a Transformer-style architecture.
15	How are images typically represented as tokens for a Transformer-based VLM?	By splitting the image into patches and embedding those patches into a sequence of vectors.
16	In fusion VLMs, where does modality fusion specifically occur?	It occurs in the cross-attention layers, where text queries interact with visual memory from the image encoder.
17	Process: Cross-Attention Fusion	Text tokens provide queries Q , while visual memory from the image encoder provides keys K and values V .
18	What is the training objective for a generative vision-language encoder-decoder?	Conditional language modeling: minimize the negative log-likelihood of predicting the next caption token given previous tokens and the image.
19	Why is a fusion VLM generally better at Visual Question Answering (VQA) than a pure dual encoder?	It can support deeper multi-step reasoning by allowing text and visual tokens to interact throughout the model.

#	Question	Answer
20	What is a primary limitation of generative fusion models compared to dual encoders?	They are computationally expensive because text generation is autoregressive during inference.
21	Concept: Generative Hallucination	A failure mode where the model produces plausible-sounding text that is not supported by the visual evidence in the image.
22	How does SigLIP's objective differ from standard CLIP?	It replaces the batch-wide softmax contrastive loss with a pairwise sigmoid loss, reducing dependence on global batch normalization.
23	Term: Distillation in VLMs	Training a small student model to mimic the embedding space and behavior of a larger pretrained teacher model.
24	What are two common loss components used to distill a TinyCLIP model?	Embedding regression using MSE loss and similarity-matrix distillation using KL divergence loss.
25	What does embedding regression represent in student-teacher distillation?	Directly minimizing the distance between the student output vectors and teacher output vectors for the same input.
26	What is the intuition behind similarity-matrix KL distillation?	The student learns to reproduce the teacher geometry by matching the pattern of similarities across a batch.
27	In a VLM decoder, what is the purpose of the causal mask?	It prevents text tokens from looking ahead at future words during training and generation.
28	How do VLMs like OWL-ViT extend CLIP for dense prediction?	They adapt image-text alignment to region-level or token-level representations for detection and localization.
29	What is Locked-image Tuning (LiT)?	A strategy where a strong pretrained image encoder is frozen while the text encoder is tuned for alignment.
30	The ability to link specific phrases in text to corresponding image regions is called what?	Grounding.
31	In the context of VLMs, what is a visual token?	A vectorized representation of a local image patch used as input to a Transformer-based encoder.
32	Why is CLIP described as an evaluator rather than a creator?	It scores how well existing text and images match; it does not synthesize new text or pixels by itself.
33	What defines the visual memory M in an encoder-decoder VLM?	It is the final sequence of contextualized visual tokens produced by the image encoder.
34	In cross-attention, what does the attention weight p_{ij} represent?	It represents how much text token i focuses on image token j .
35	What is teacher forcing in the context of VLM training?	Using ground-truth caption tokens as decoder inputs instead of the model's own previous predictions.
36	Why might a dual encoder struggle with compositional counting tasks?	Global alignment compresses the image into one vector and may lose fine-grained spatial and counting details.
37	How is image-to-text retrieval performed using CLIP?	Embed the query image and retrieve text embeddings with the highest cosine similarity scores.
38	Concept: Hard Negatives	Mismatched image-text pairs that the model currently predicts as highly similar and must push apart strongly.
39	In the student-teacher training loop, why is the teacher model set to eval() mode?	To make teacher embeddings consistent and to disable training-specific behavior such as dropout.
40	What is the intuition behind using BPE tokenizers in VLMs like CLIP?	They handle subword units, represent rare words, and keep the vocabulary size fixed.
41	How does CoOp (Context Optimization) improve VLM performance?	It replaces hand-crafted prompts with continuous learnable prompt vectors optimized for a specific task.

#	Question	Answer
42	What is the modality gap in vision-language embedding spaces?	The tendency for image and text embeddings to occupy different regions even when they are semantically aligned.
43	In fusion models, when is the FFN usually applied?	After the attention mechanisms, to process each token representation independently.
44	Term: BLIP-2	A model that uses a Q-Former to bridge a frozen image encoder and a frozen large language model.
45	What does greedy decoding mean in caption generation?	Selecting the token with the highest predicted probability at each generation step.
46	Why does CLIP scale the dot product by $1/\tau$?	To transform raw similarities into scaled logits so the softmax gradients have an effective learning scale.
47	In generative VLMs, what does the token <bos> indicate?	The beginning of a sequence.
48	What is the primary motivation for TinyCLIP or MobileCLIP?	Reducing the accuracy-latency-memory tradeoff so VLMs can run efficiently on edge or mobile devices.
49	What is the significance of compositional supervision in VLM training?	Language provides flexible descriptive labels instead of static fixed categories.
50	How does image-text contrastive learning differ from image-text matching (ITM)?	Contrastive learning aligns embeddings globally, while ITM often performs binary classification of whether a pair matches.
51	In CLIP, if the logit scale is 14.0, what is the corresponding temperature τ ?	$\tau \approx 1/14.0$, because logit scale is the inverse of temperature.
52	Formula: Cosine similarity between vectors A and B	$\cos(A,B) = A^T B / (\ A\ _2 \ B\ _2)$.
53	What happens if two embeddings are exactly the same in a unit-normalized space?	Their cosine similarity is 1.0, the maximum possible value.
54	In distillation, what is the goal of affinity mimicking?	To make the student capture the same relative relationships between data points as the teacher.
55	What is a patch embedding?	A linear projection of a flattened 2D image patch into a 1D vector compatible with Transformer input.
56	Why is web-scale data for VLM training often called noisy?	Because alt-text or metadata can contain irrelevant, biased, stereotyped, or inaccurate descriptions of the paired image.
57	Retrieving the top-k items from a large embedding database is often accelerated by what search?	Approximate Nearest Neighbor (ANN) search.
58	How does a visual instruction-tuned model like LLaVA differ from standard CLIP?	It is fine-tuned on dialogue-style data to follow natural-language instructions about an image.
59	In the CLIP loss, what does B usually represent?	Batch size.
60	What does a t-SNE visualization help demonstrate in VLM analysis?	It shows whether semantically similar images and text prompts cluster together in a 2D projection of latent space.
61	Why is prompt engineering critical for CLIP zero-shot performance?	Different prompt phrasing changes the text embedding and can affect alignment with image features.
62	In distillation training, what is the student logit scale?	A learnable student parameter that scales similarity scores, mirroring temperature scaling.
63	How do VLMs use global alignment?	They match the entire image and the entire sentence as global units rather than focusing only on internal parts.
64	Concept: Multimodal Fusion	Integrating features from different data types, such as image and text, into a single representation or processing stream.

#	Question	Answer
65	What is the role of the LM head in an encoder-decoder VLM?	It projects decoder hidden states to the vocabulary size to produce next-token probabilities.